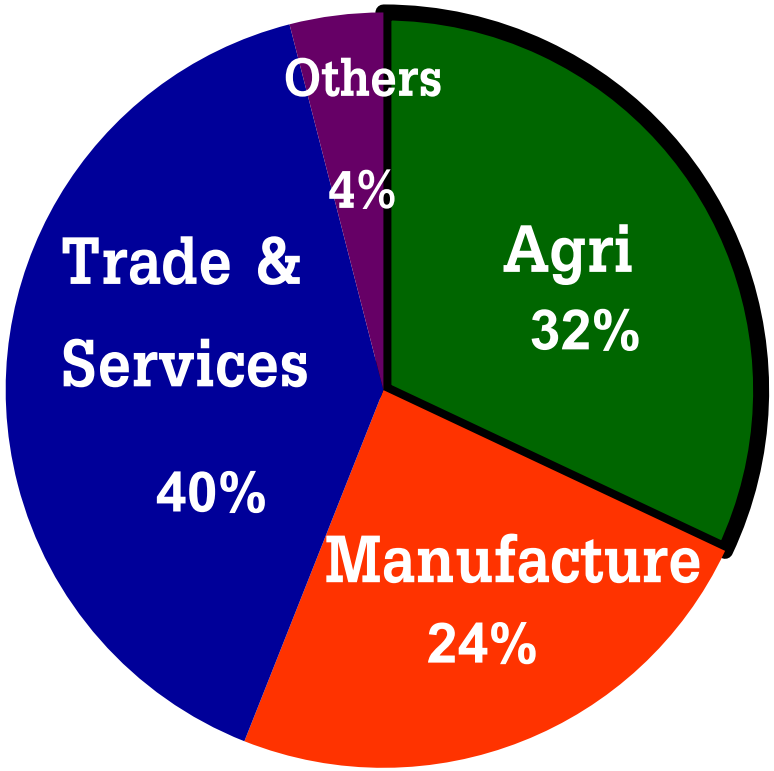
Modern Farm Management Technologies and Their Adoption

Isara Vongkusolkit, Chairman of Mitr Phol Group

Imbalance of Income Generated from Agricultural Sector

Employed



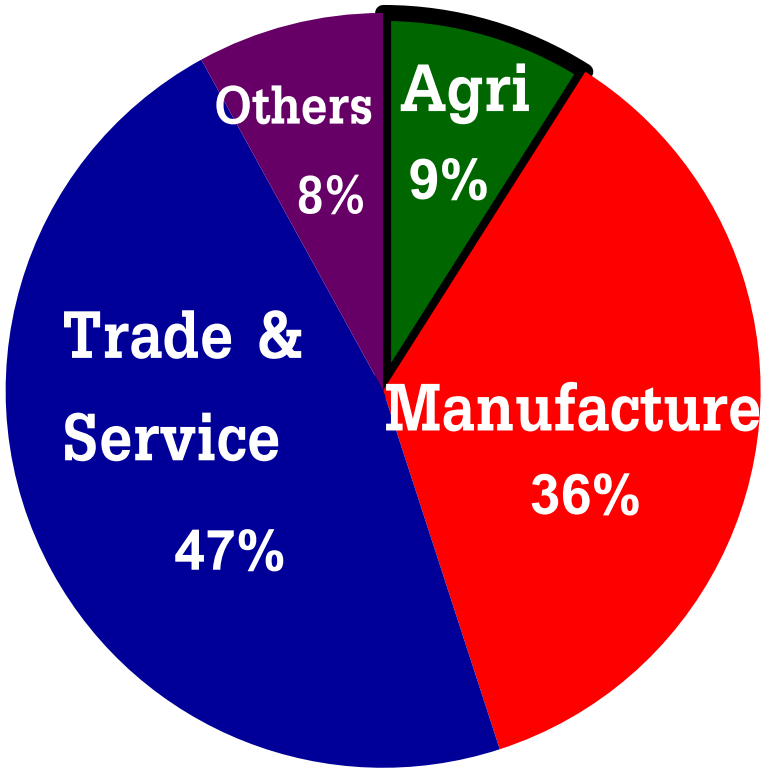
12 million farmers or

32% of total labor force

generate income only

9% of GDP

% of GDP



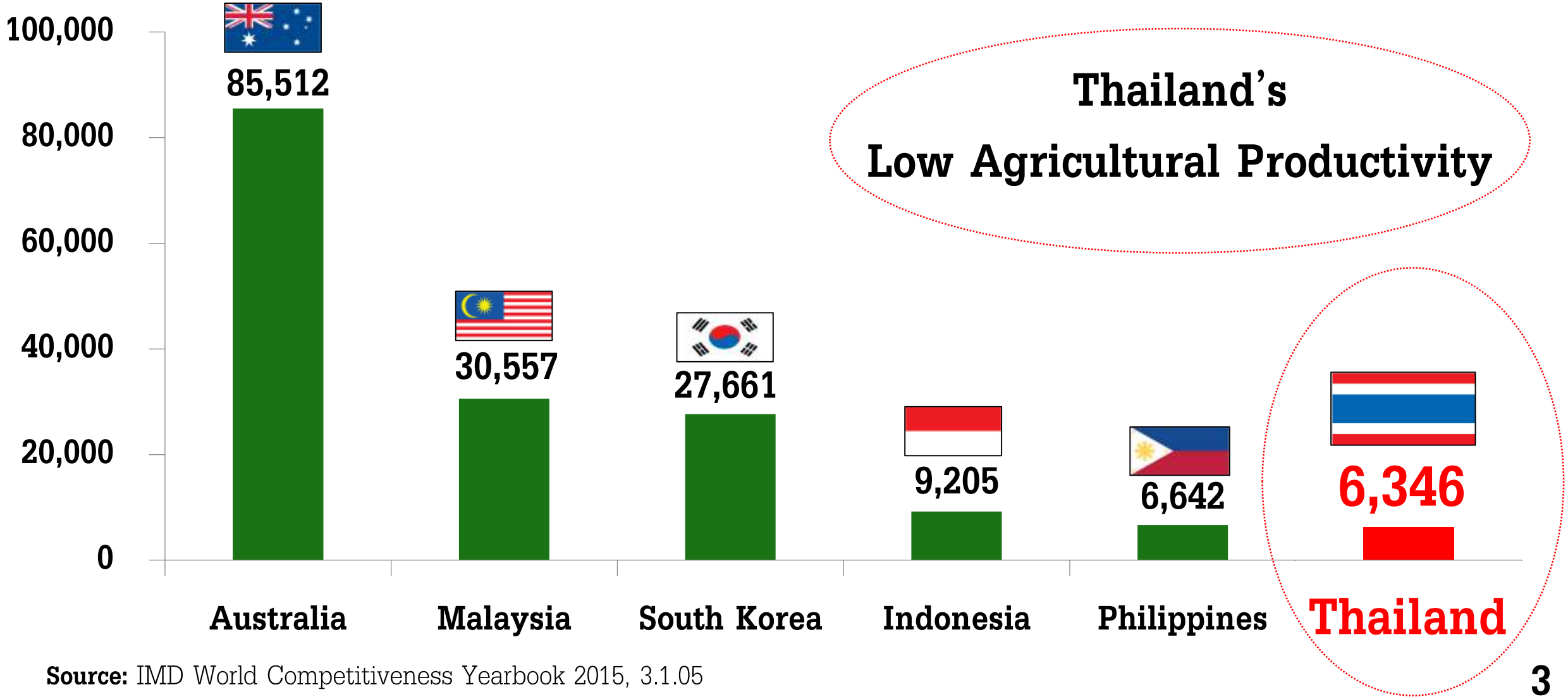
Malaysia, 1.65 million farmers or 11% of total labor force generate income 8% of GDP

Australia, 0.52 million farmers or 4% of total labor force generate income 4% of GDP

(Source : National Statistical Office, Bank of Thailand (2015), World Fact book)

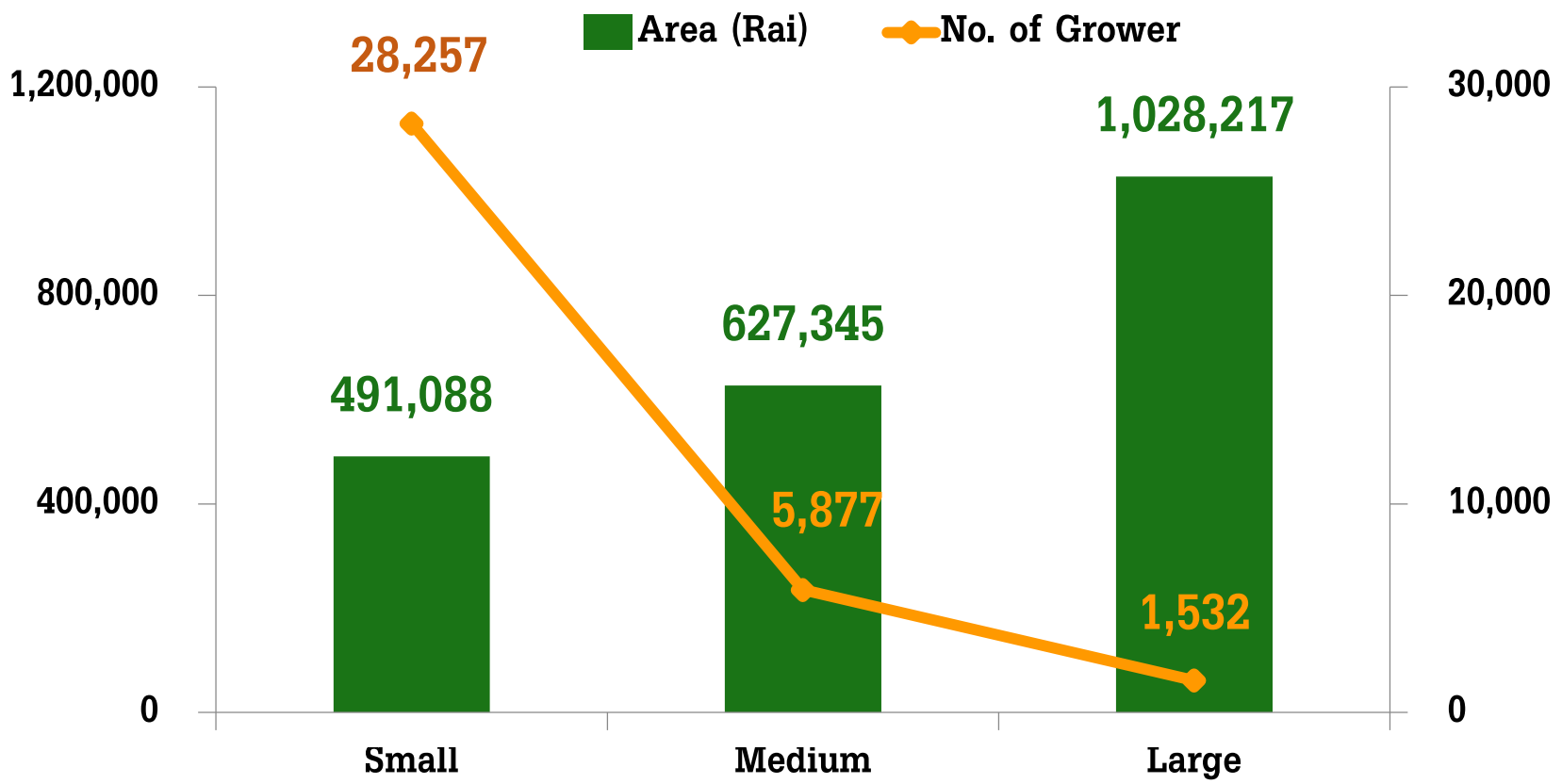
Agricultural Productivity

Unit : US$/Person employed/Year



Source: IMD World Competitiveness Yearbook 2015, 3.1.05

Cane Grower Structure



Total Growers : 35,666 family

Total Cane Area : 2,146,650 rai

Small-Scale Grower : area $\leq$ 50 rai (<8 Ha)

Medium-Scale Grower : area 51-250 rai (8-40 Ha)

Large-Scale Grower : area $>$ 250 rai (>40 Ha)

Cane Extension System

1. Production Factors & Finance – Short Term Loan (1 Year)
2. Irrigation and Equipment – Medium-term Loans (3 Years)
3. Mechanization - Medium – Long term Loans (3-5 Years)

Budget: 7,774 MB per year, covering 35,666 farmer's family

Small-Scale Farmer Development (<50 Rai or < 8 Ha)

Construction of Cane Loading Station to Reduce Cost of Transportation

Cost of harvesting and transportation in 60 km distance (Baht/Ton)

Expenses	Sugar Mill	Loading Stations
Cane Cutting	130	-
Loading	70	-
Transportation	150	125
Fuels	-	20
Total	350	145

Cost of transportation could be reduced by 205 Baht/ton or 2,050 Baht/Rai or equivalence to 41,000 Baht/family

Sample of Logistics Improvement in China Operations

MIPENG 蜜朋

Not pass the cane Loading station

Delivery time: 37 hr.

The hours of transferring to factory

87.5 hr.

Operation time: 8.8 hr.

Preparing the truck: 41.7 hr.

Pass the cane Loading station

Harvest the canes then inform the factory: 12.7 hr.

The hours of transferring to factory

31.3 hr.

Manage the truck to factory: 10.3 hr.

Preparing the truck: 8.2 hr.

Water Usage Efficiency

Increase yield from 8.90 ton/rai to 15.00 ton/rai

Mitr Phol Research and Development Institute

Establish in 1996

Objective : To develop sugarcane's productivity and farm management. In addition, to transfer know-how for farmers in order to make them have a sustainable career

Annual R&D Budget : 450 Million Baht

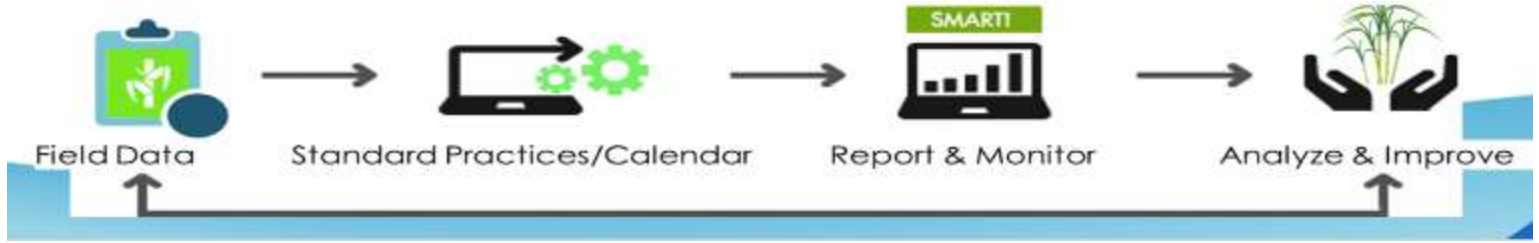
(about 0.5% of revenue)

Mitr Phol "Modern Farm"

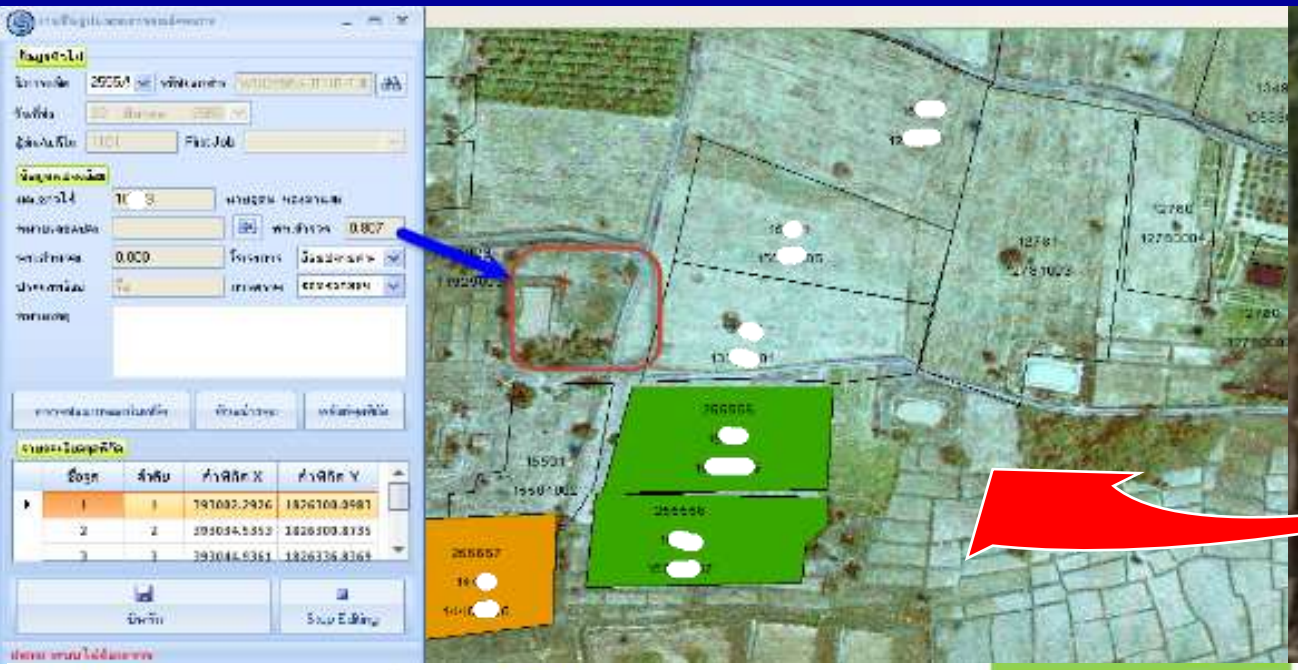
- Innovation, Science and Technology, R&D
- Precision Farming
- Minimize risk of uncontrolled factors i.e. weather, rain, and etc
- Mechanization and Utilization
- Combination of production i.e. Cooperatives
- Standardize, Traceability and Benchmarking

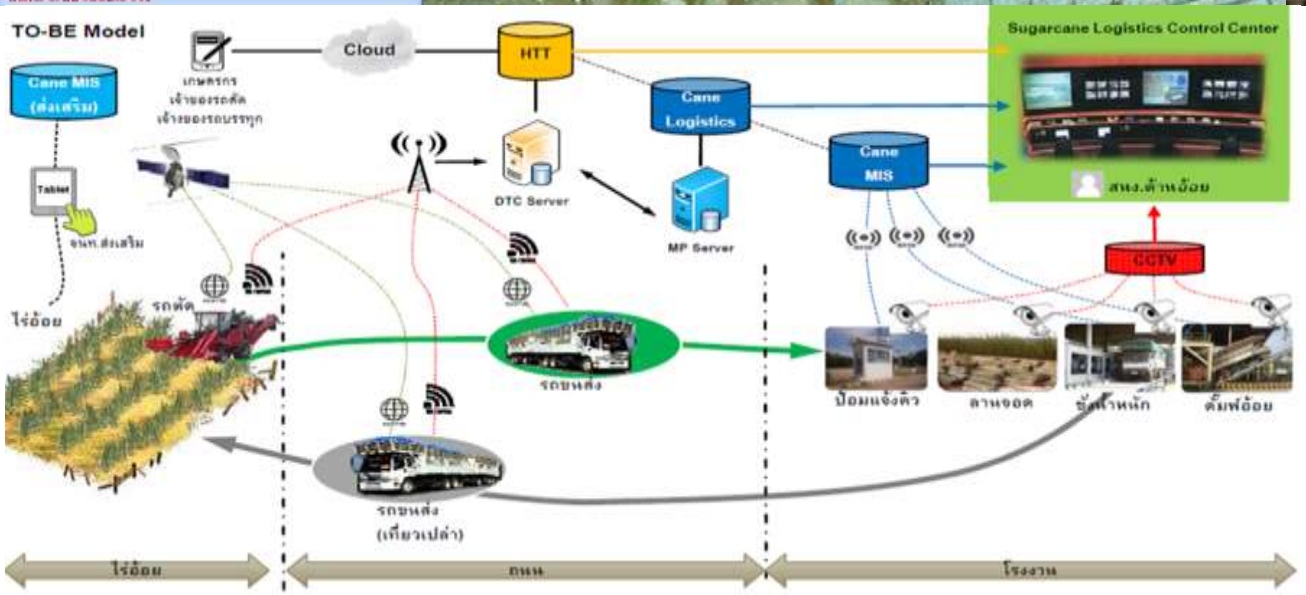
Land Leveling | Precision Guidance | Sugarcane Planting | Cultivation | Yield Mapping

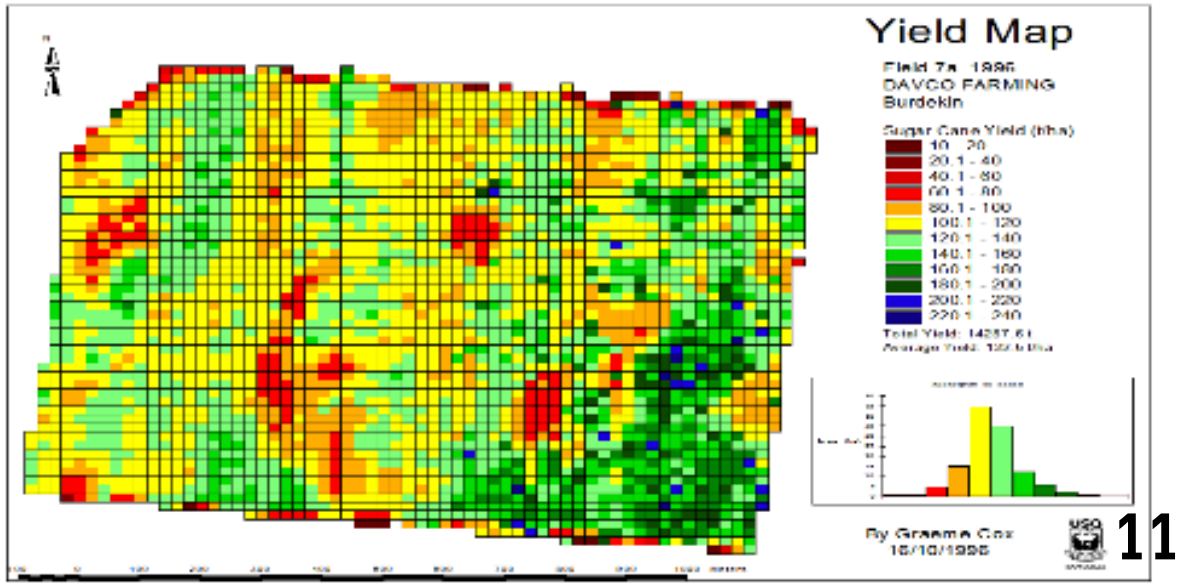
- Increase Competitiveness
- Increase Yield / Reduce Cost
- Reduce Environmental Impact



Precision Farming : IT Application

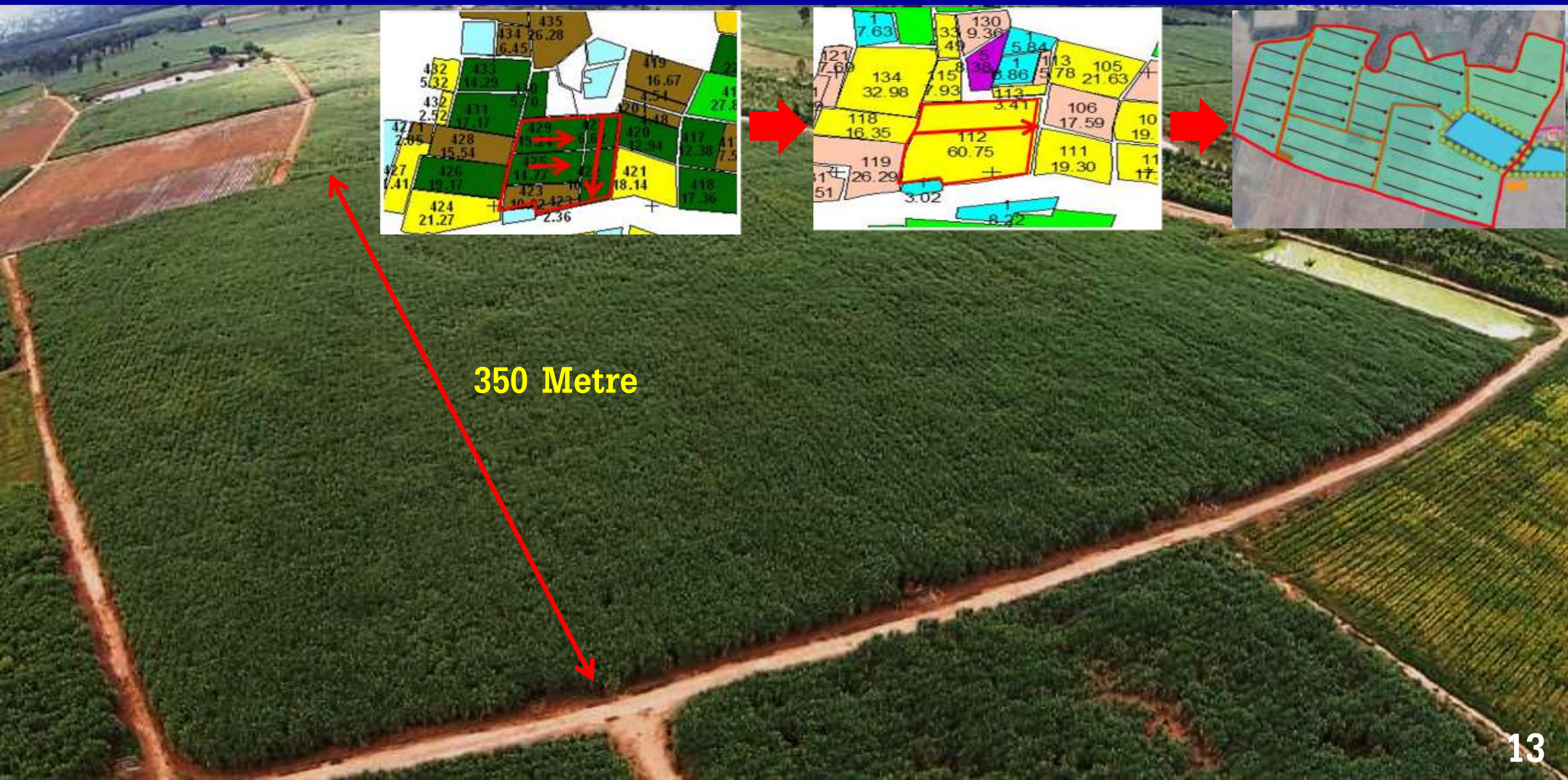






Mechanization: Increase production capacity and solve labour shortage

Land Preparation and Consolidation to be Large Scale Farming



Efficiency Comparison of Harvester and Labor

Australia		Thailand	
▪ 1 Harvester	750 tons/day	▪ Cane Cutting	750 tons/day
▪ Labor	6 heads	▪ Labor	620 heads

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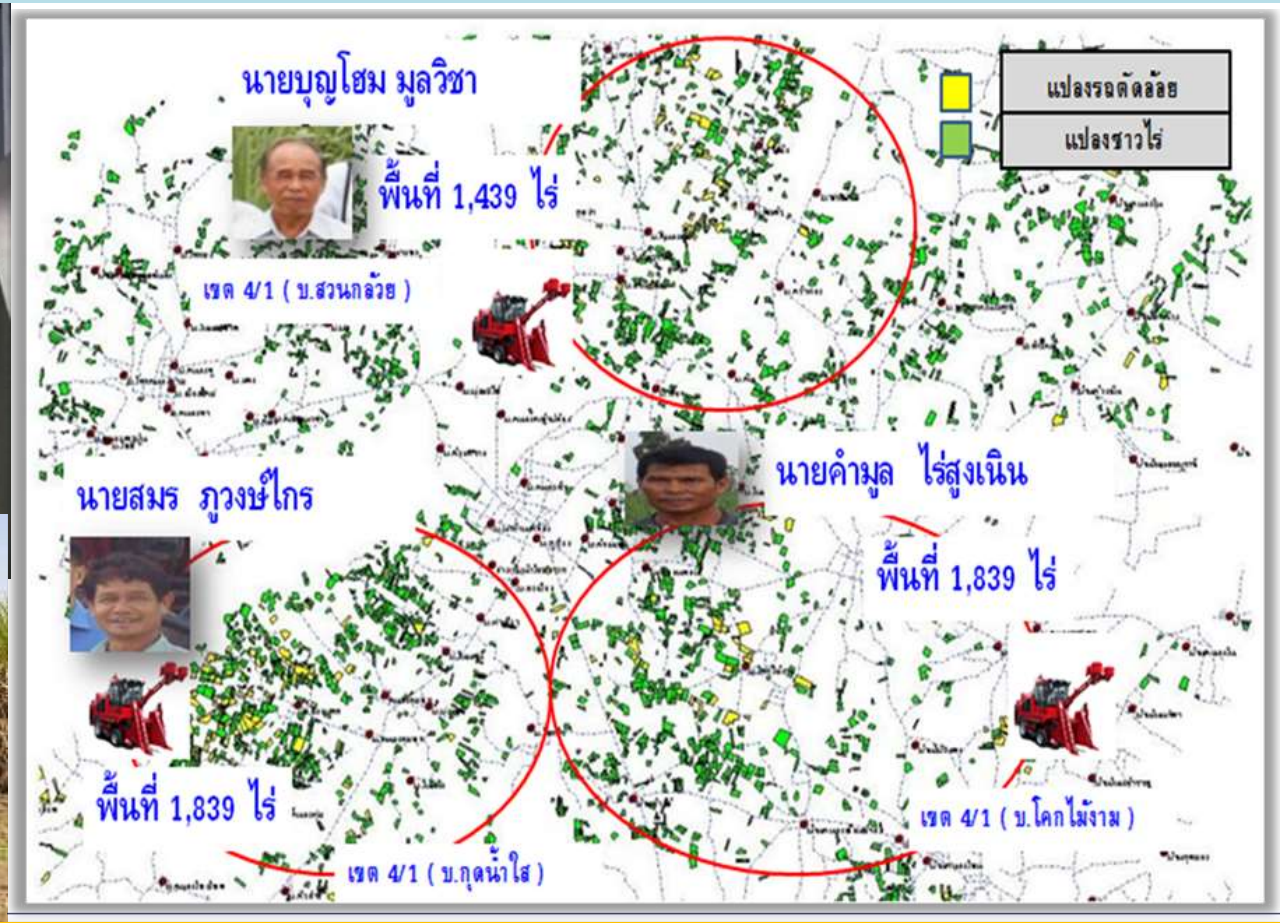
Large-Scale Farmer Development (>250 Rai or >40 Ha)

Comparison of Cane Cutting : Labor and Harvester

Items	Cost (Baht/Ton)	
	Labor	Harvester
Cutting + Loading	200	180
Transport	120	100
Total	320	280
Diff	40	

Large-Scale Farmer Development (>250 Rai or >40 Ha)

Farm Mechanization



205 Farmer Groups who use harvester for cutting 1.27 Mil.tons of cane,

Cost Saving of 51 Mil.Baht a year

Cane Farmer Successor Program

To promote new generation of farmer family to back their hometown

- **Training "Cane Farmer Successor Program" and Site Visit "ModernFarm" in Australia**
- **Use mechanization and IT to be implemented in farm**
- **Use dripping irrigation and solar-cell water pump instead of traditional ways**

Personnel Development: Bilateral Agreement

GERMAN-THAI
Dual Excellence Education

MITR PHOL GROUP

Collaboration of Vocational Education Commission, German-Thai Chamber of Commerce and Mitr Phol

- To solve shortage of skill labor in sugar and bio energy industry
- To develop teacher to have knowledge and skill in sugar and bio energy industry

Education	No. of Scholarship
Electrical	93
Plant Science	80
Mechanical	74
Agri-Mechanical	69
Mechatronic	38
Production Technology	19
TOTAL	373

Total Students in Program: 373 Students

Vocational Education, Modern Farm

Vocational Education – “Dual-education system”

Modern Farm Training

Promote Sufficiency Economy Philosophy of King Rama IX

Cash crop can be reduced household expenditures of 12,000 Bt/household/year

MITR PHOL
Sugar

Sustainable Model :
"GROW TOGETHER"